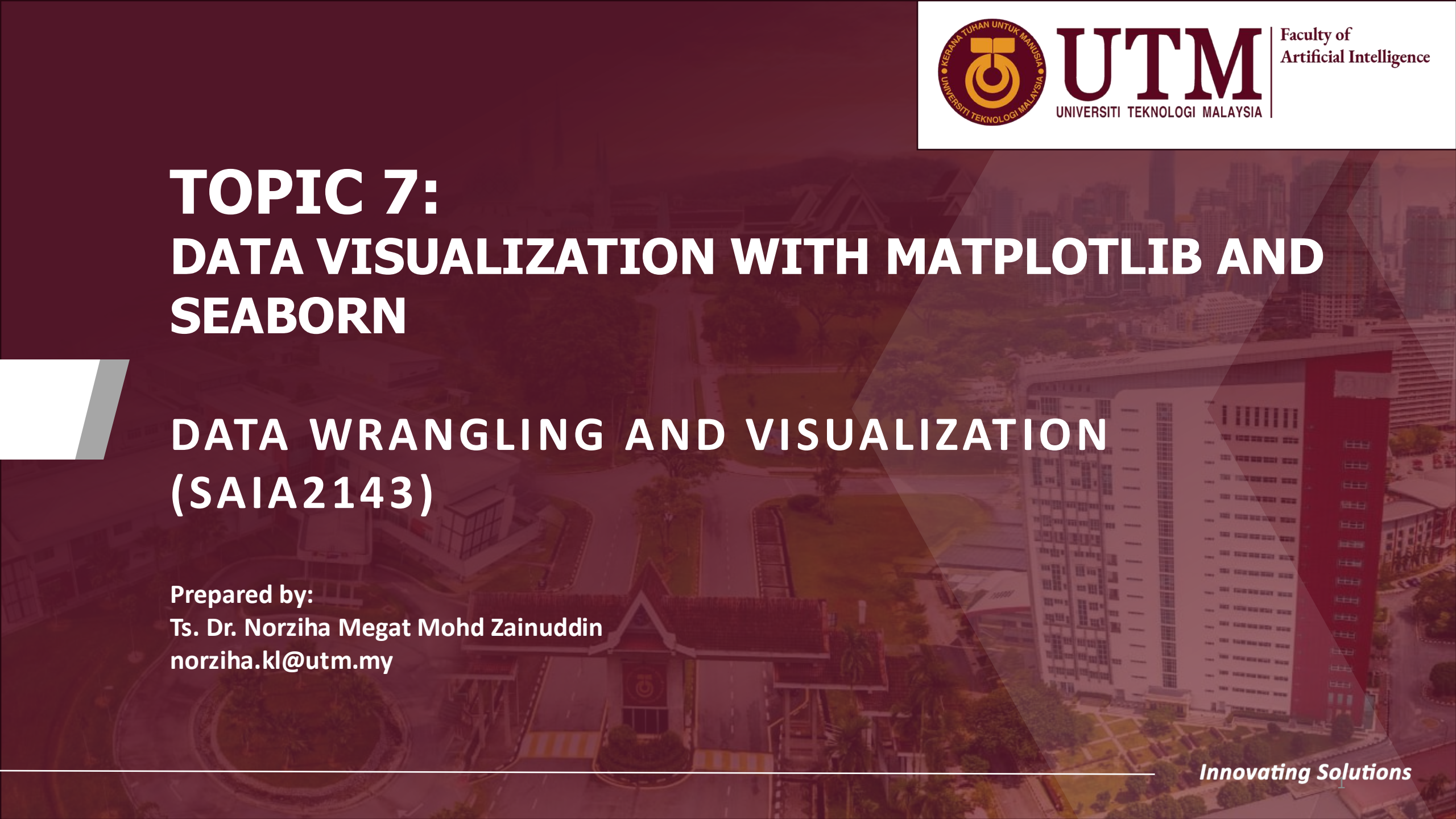


UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 7:
DATA VISUALIZATION WITH MATPLOTLIB AND SEABORN

DATA WRANGLING AND VISUALIZATION (SAIA2143)

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COURSE LEARNING OUTCOMES

At the end of this two hours class, students should be able to:

a. Identify an overview, anatomy and types of plots in Matplotlib for Python.
b. Creating various charts and plot in Matplotlib.
c. Creating attractive and informative statistical graphics using Seaborn.
d. Customizing color pallets and themes for visual appeal.

TOPIC 7: DATA VISUALIZATION WITH MATPLOTLIB AND SEABORN

a. **Overview of Matplotlib for Python**
b. Creating various plots: line charts, bar charts, scatter plots
c. Introduction to the Seaborn library for statistical data visualization

Introduction to matplotlib

a. What is Matplotlib?

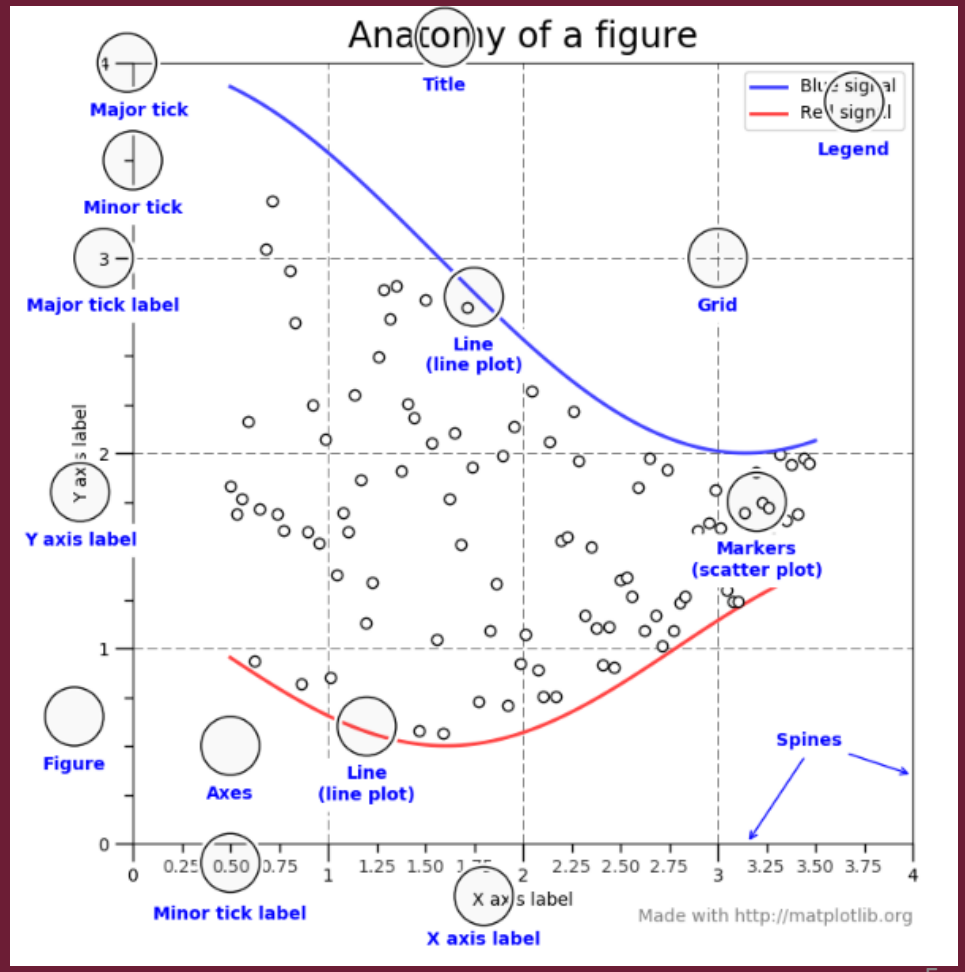
- A comprehensive library for creating static, animated, and interactive visualizations in Python.
- Developed by John D. Hunter in 2003.

b. Why use Matplotlib?

- Highly customizable
- Widely used in the scientific and data analysis communities
- Supports multiple backends and output formats

Anatomy of a Matplotlib Plot

- **Figure**: The entire window or image
- **Axes**: The area where data is plotted
- **Axis**: X and Y axes within the Axes
- **Other components**:
  - **Title**: Description of the plot
  - **Labels**: Text for X and Y axes
  - **Ticks**: Markings on the axes
  - **Legend**: Explains plotted elements



Common Types of Plots in Matplotlib

- **Line Plot**: Shows trends over time or ordered categories
- **Bar Chart**: Compares quantities across categories
- **Scatter Plot**: Shows relationship between two variables
- **Histogram**: Displays distribution of a dataset
- **Boxplot**: Summarizes data distribution using quartiles

Basic Plotting with pyplot

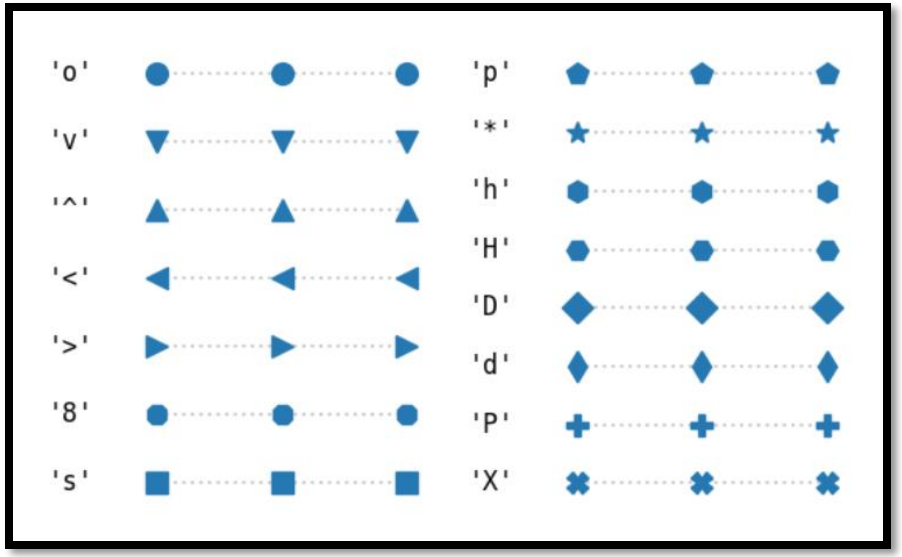
```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4]
y = [10, 20, 25, 30]
plt.plot(x, y)
plt.title("Basic Line Plot")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

Customizing Plots in Matplotlib

- **Titles and Labels**:
  - `plt.title("Title"), plt.xlabel("X"), plt.ylabel("Y")`

- **Line Styles and Colors**:
  - Line: `'-', '--', '-.'`
  - Color: `'r', 'g', 'b',` or hex codes

- **Markers**:
  - `'o', 's', '^', '*', 'x'`



Reference



Instruction:

Interested about other markers?
Let's visit this website.

https://matplotlib.org/3.1.0/gallery/lines_bars_and_markers/marker_reference.html

Customizing Plots in Matplotlib

- **Legend and Grid**:

```
plt.legend(["Line 1"])
plt.grid(True)
```

- **Figure Size and Style**:
  - `plt.figure(figsize=(8,6))`
  - `plt.style.use('ggplot')`

TOPIC 7: DATA VISUALIZATION WITH MATPLOTLIB AND SEABORN

a. Overview of Matplotlib for Python

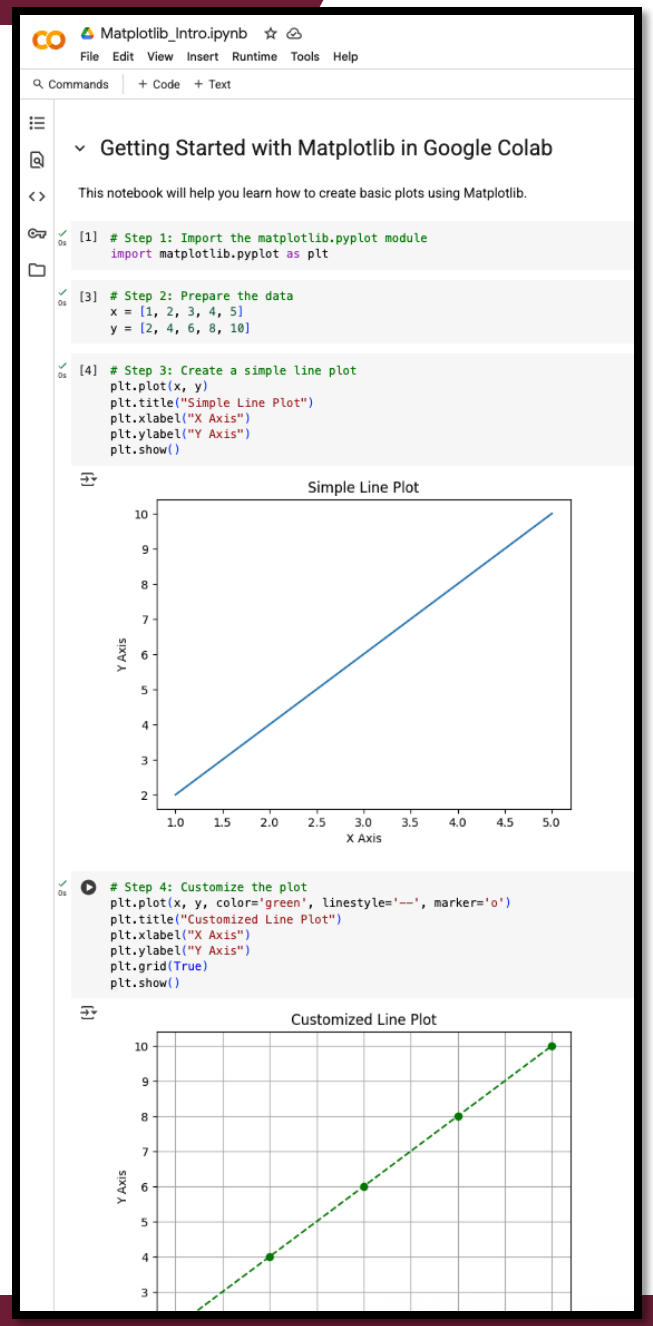
b. Creating various plots: line charts, bar charts, scatter plots

c. Introduction to the Seaborn library for statistical data visualization

Creating Line Charts

- Used for showing trends or changes over time
- Example:

```
x = [0, 1, 2, 3]
y = [0, 1, 4, 9]
plt.plot(x, y, marker='o', linestyle='-', color='blue')
plt.title("Line Chart")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()
```



CLASS ACTIVITY

Instruction:

1. Run the file
2. Student Exercise: Plot the graph using the following data:
   x = [0, 1, 2, 3, 4, 5]
   y = [0, 1, 4, 9, 16, 25]
   - Customize the color to blue
   - Use a dashed line ('--')
   - Use a square marker ('s')
3. What kind of graph show?

Google Colab file:
Matplotlib_Intro.ipynb

www.fai.utm.my

Creating Bar Charts

- Ideal for comparing different categories
- Horizontal bars: `plt.barh()`
- Example:

```
categories = ['A', 'B', 'C']
values = [5, 7, 3]
plt.bar(categories, values, color='orange')
plt.title("Bar Chart")
plt.xlabel("Category")
plt.ylabel("Value")
plt.show()
```

Creating Scatter Plots

- Best for showing relationships between two numeric variables.
- Useful in correlation analysis and outlier detection
- Example:

```
x = [1, 2, 3, 4, 5]
y = [5, 4, 3, 2, 1]
plt.scatter(x, y, color='red', marker='^')
plt.title("Scatter Plot")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()
```

CLASS ACTIVITY

Instructions:

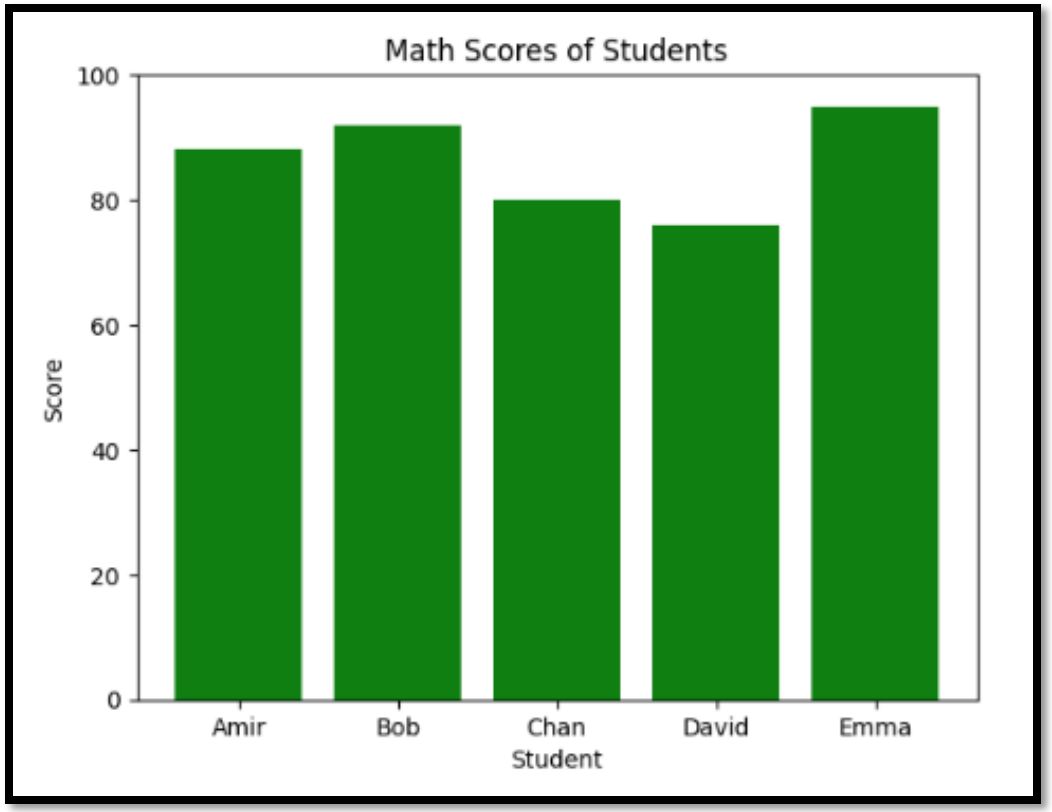
Part 1: Bar Graph – Math Scores

1. Plot a bar graph to show **math scores** of each student.
2. Use **student names** on the x-axis and **scores** on the y-axis.

```
# Dataset
students = ["Amir", "Bob", "Chan", "David", "Emma"]
math_scores = [88, 92, 80, 76, 95]
science_scores = [85, 89, 82, 78, 90]
```

3. Add a **title**, **axis labels**, and change the **bar color** to green.

Output:



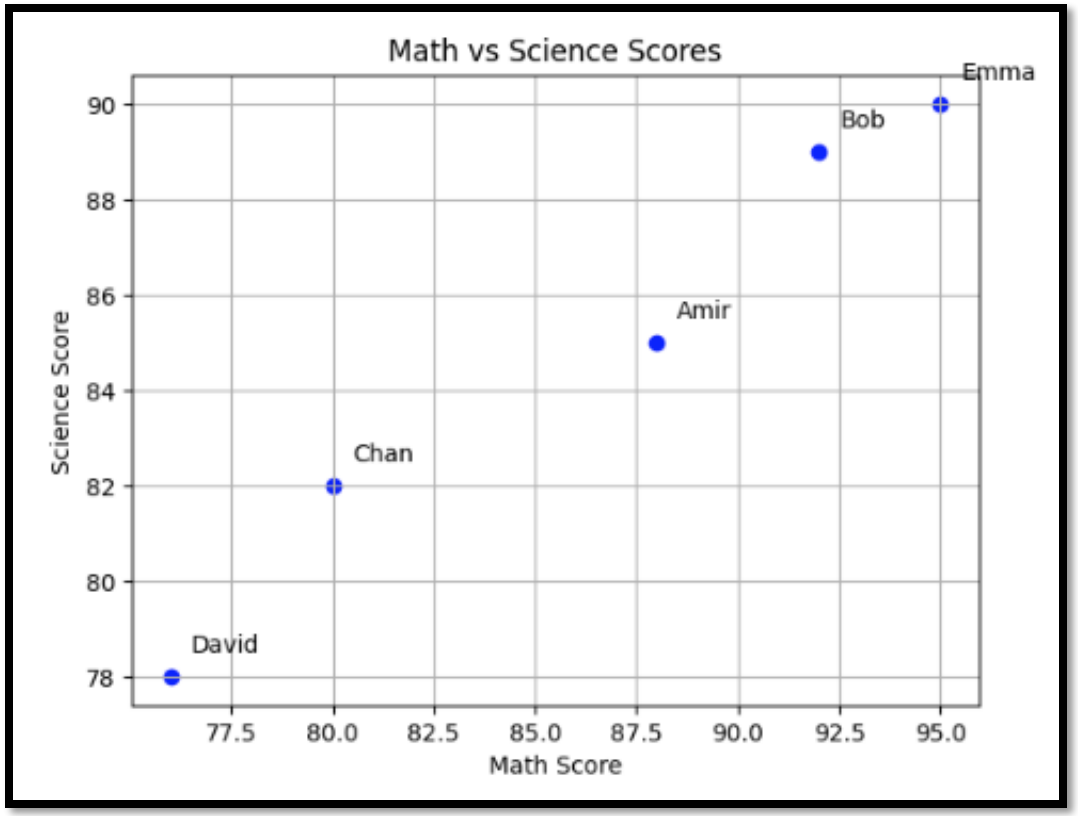
CLASS ACTIVITY

Instructions:

Part 2: Scatter Plot – Math vs Science Scores

1. Create a scatter plot comparing **math vs. science scores**.
2. Plot math scores on the x-axis and science scores on the y-axis.
3. Add **student names as annotations** on each point.

Output:



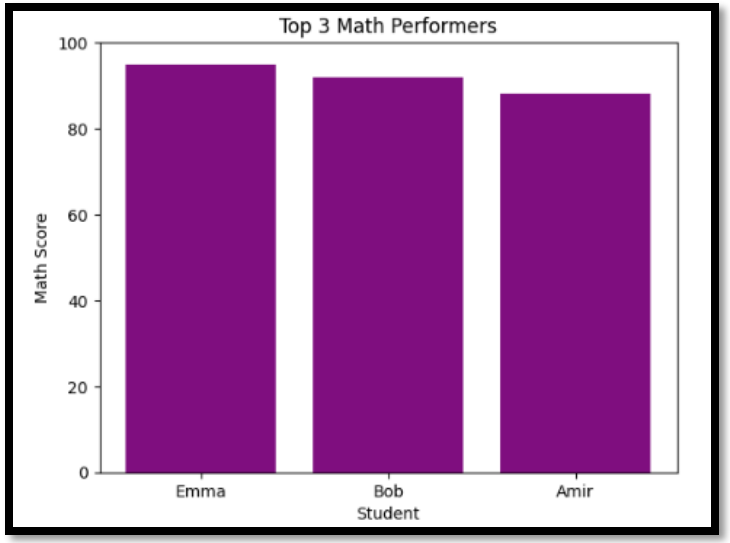
CLASS ACTIVITY

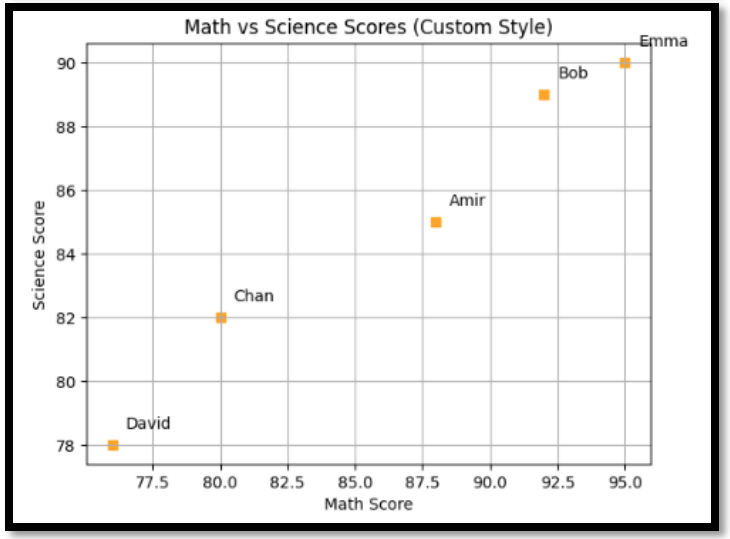
Instructions:

Part 3:

a. Sort students by math scores and plot the **top 3 performers** using a new bar chart.
b. Change the **marker shape** and **color** in the scatter plot.

Output:





TOPIC 7: DATA VISUALIZATION WITH MATPLOTLIB AND SEABORN

a. Overview of Matplotlib for Python

b. Creating various plots: line charts, bar charts, scatter plots

c. **Introduction to the Seaborn library for statistical data visualization**

Introduction to seaborn

- Built on top of Matplotlib
- Provides high-level interface for drawing attractive and informative statistical graphics
- Integrates well with Pandas DataFrames
- Developed for statistical data visualization

https://seaborn.pydata.org/tutorial/introduction.html

www.fai.utm.my

Why Use Seaborn?

- Simpler syntax than Matplotlib for common visualizations
- Built-in themes and color palettes
- Support for complex plots (e.g., heatmaps, violin plots) with minimal code
- Useful for exploring and understanding data relationships
- Automatically interprets column names with Pandas DataFrames
- Supports grouping and categorical data handling

Installing and Importing Seaborn

- Installation:

```
pip install seaborn
```

- Import:

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Seaborn vs Matplotlib

Feature	Matplotlib	Seaborn
Interface	Low-level	High-level
Plot aesthetics	Manual	Automatic styles
Data structure	Arrays	DataFrames
Plot types	Basic	Statistical

Example: Basic Scatter Plot with Seaborn

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

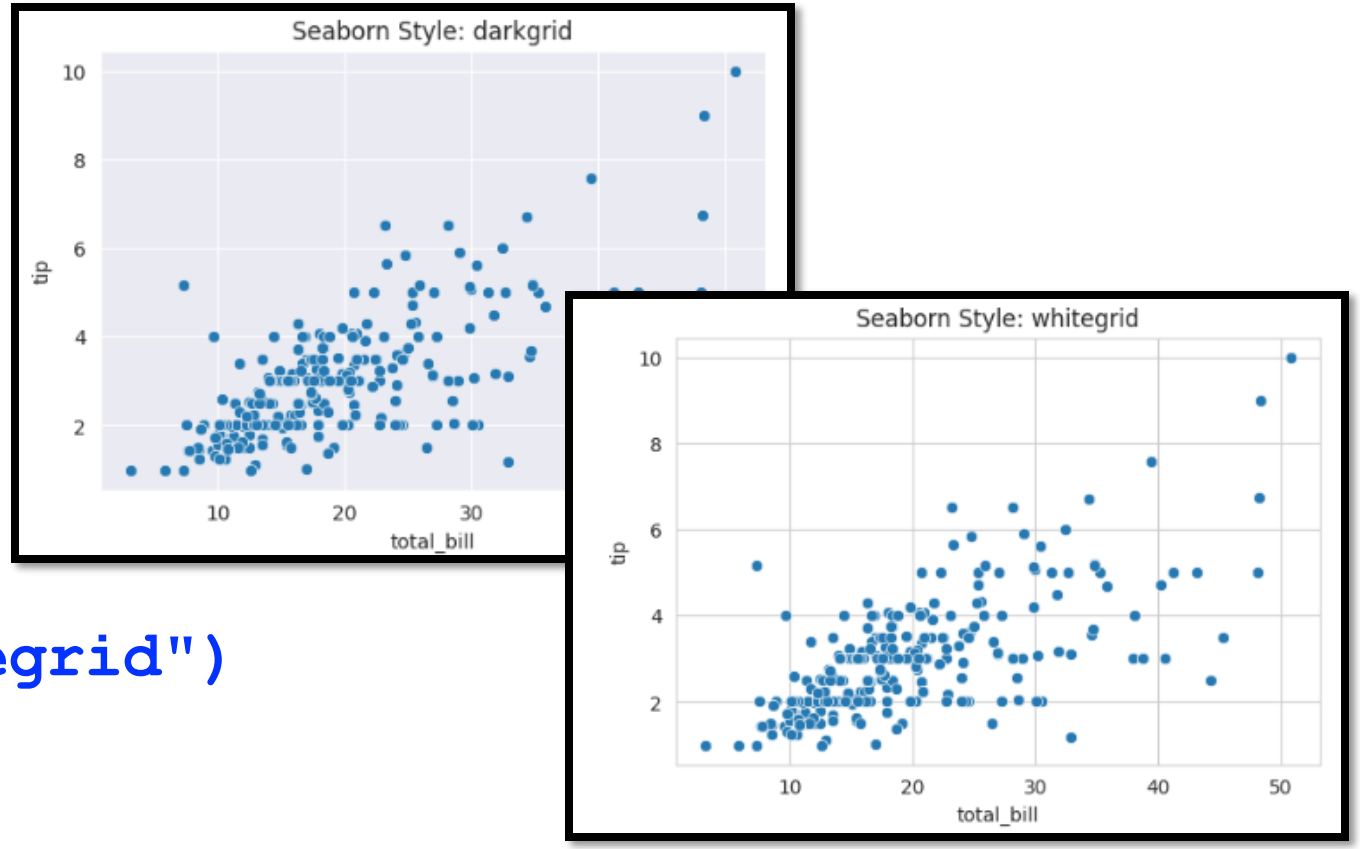
data = sns.load_dataset("tips")
sns.scatterplot(x="total_bill", y="tip", data=data)
plt.show()
```

Seaborn Themes and Styles

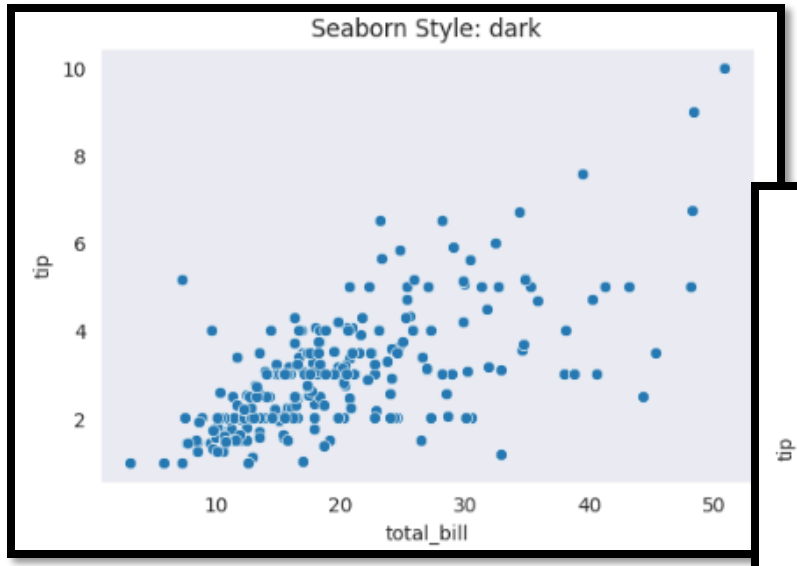
- Five built-in themes:
  i. darkgrid (default)
  ii. whitegrid
  iii. dark
  iv. white
  v. ticks

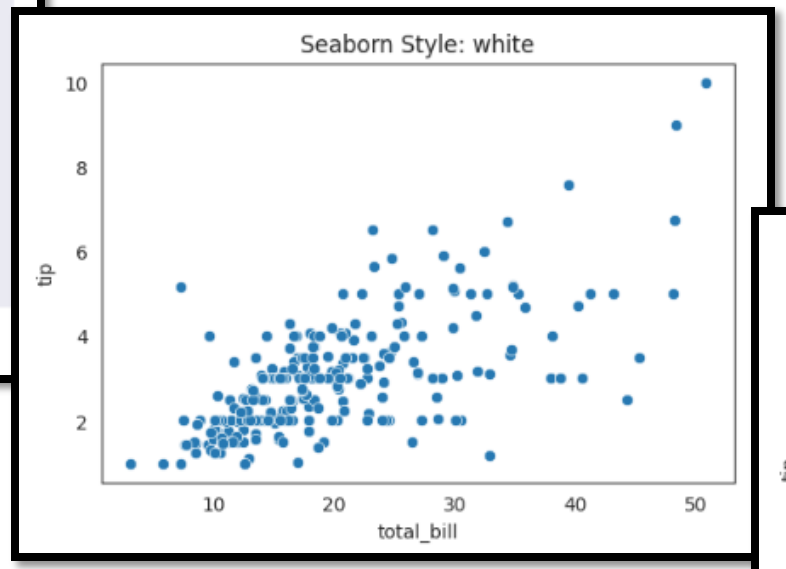
Example:

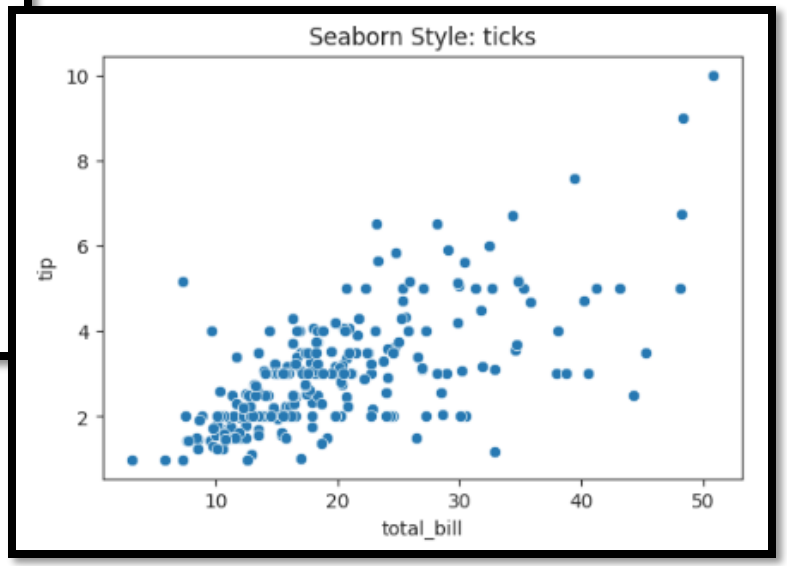
```
sns.set_style("whitegrid")
```



Seaborn Themes and Styles

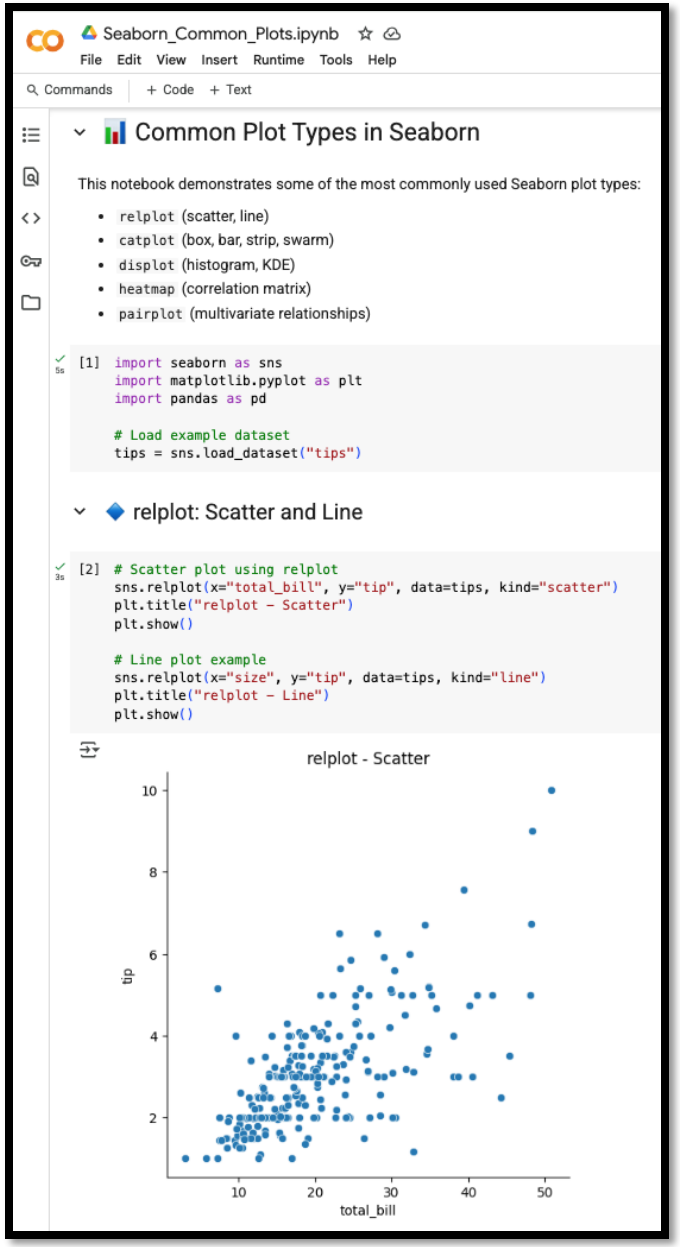






Common Plot Types in Seaborn

- **relplot** (scatter, line)
- **catplot** (box, bar, strip, swarm)
- **distplot** / **displot** (histograms, KDE)
- **heatmap** (correlation matrices)
- **pairplot** (multivariate relationships)



Instruction:

1. Run the file and recognize the most common Seaborn plot types

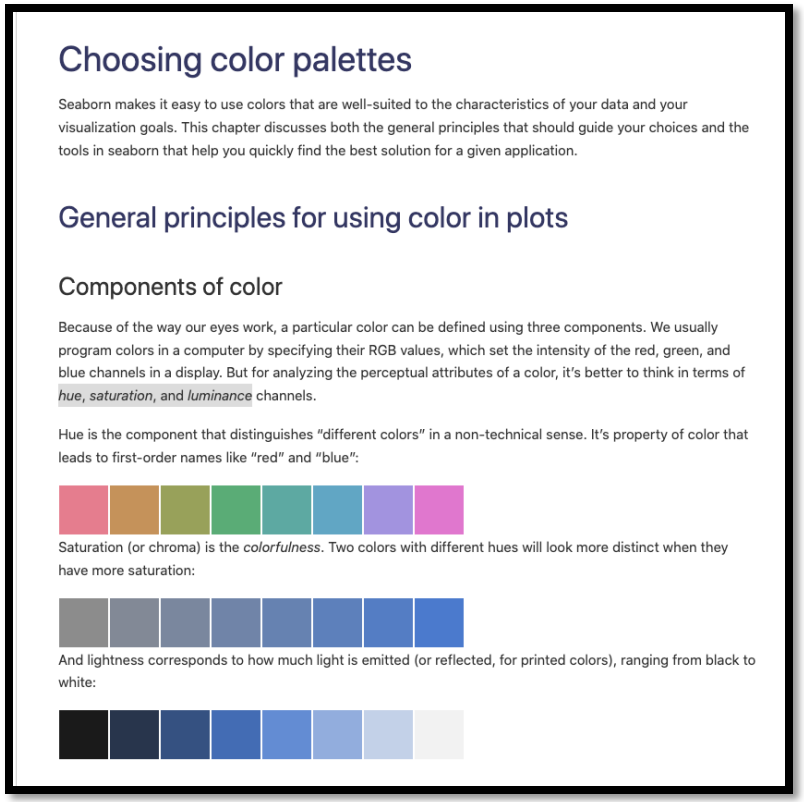
Google Colab file:

Seaborn_Common_Plots.ipynb

Seaborn Color Palettes

- Seaborn offers built-in palettes for visual appeal
- Can be categorical, sequential, or diverging
- Integrates smoothly into all Seaborn plots

CLASS ACTIVITY



Choosing color palettes

Seaborn makes it easy to use colors that are well-suited to the characteristics of your data and your visualization goals. This chapter discusses both the general principles that should guide your choices and the tools in seaborn that help you quickly find the best solution for a given application.

General principles for using color in plots

Components of color

Because of the way our eyes work, a particular color can be defined using three components. We usually program colors in a computer by specifying their RGB values, which set the intensity of the red, green, and blue channels in a display. But for analyzing the perceptual attributes of a color, it's better to think in terms of *hue*, *saturation*, and *luminance* channels.

Hue is the component that distinguishes "different colors" in a non-technical sense. It's property of color that leads to first-order names like "red" and "blue":

Saturation (or chroma) is the *colorfulness*. Two colors with different hues will look more distinct when they have more saturation:

And lightness corresponds to how much light is emitted (or reflected, for printed colors), ranging from black to white:

Seaborn Color Palettes Overview

1. Let's visit this website Choosing Color palettes in Seaborn.

   https://seaborn.pydata.org/tutorial/color_palettes.html

   What is the different

Matplotlib Color Palettes

- `cmap` is a colormap library for python, providing all of the colormaps in matplotlib.
- Recognized Color Names :

https://cmap-docs.readthedocs.io/en/latest/colors/



When to Use Which Palette?

- **Categorical data**: Deep, Muted, Bright, Pastel, Dark, Colorblind
- **Sequential data**: Blues, Greens, Reds, Purples
- **Diverging comparisons**: coolwarm (blue-white-red gradient), RdBu (red-white-blue gradient)
- Example: `sns.set_palette("pastel")`

Customizing Color Palettes

- Use `sns.color_palette()` to create a custom palette
- Example:

```
custom_palette = sns.color_palette(["#FF6347", "#4682B4",
"#FFD700"])
sns.set_palette(custom_palette)
```

- Works with hex codes and RGB

Previewing Palettes

- Use sns.color_palette("palette_name") to preview
- Display with sns.palplot()
- Example:

```
sns.palplot(sns.color_palette("bright"))
```

Combining Themes and Palettes

- Example:

```
sns.set(style="white", palette="pastel")
sns.boxplot(x="day", y="total_bill", data=tips)
```

- Enhances plot elegance and clarity

CLASS ACTIVITY

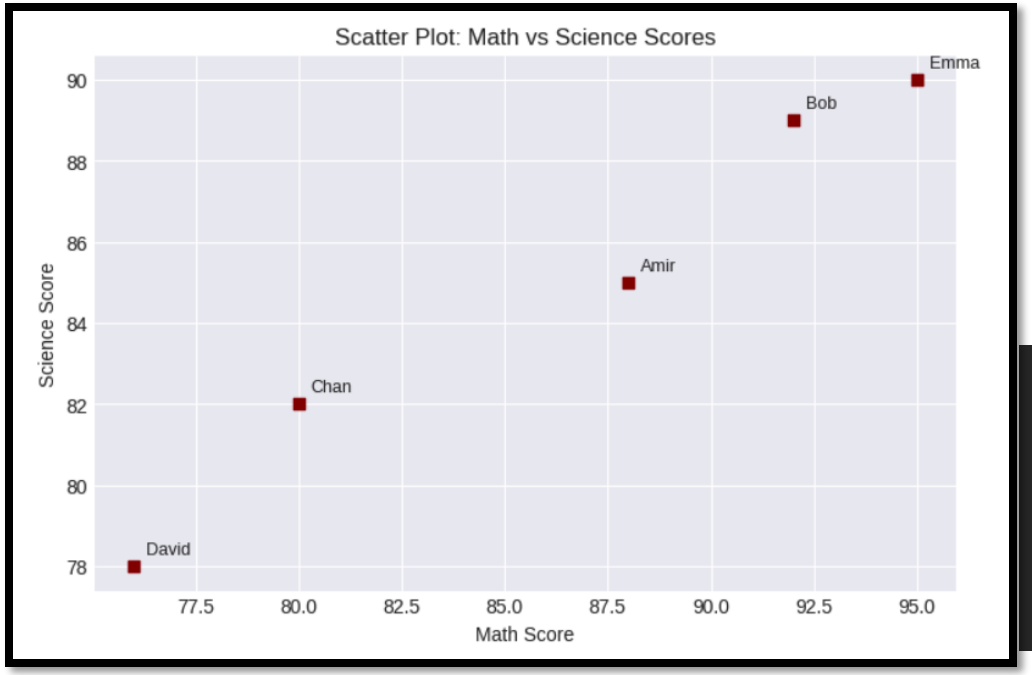
Instructions:

Part 4:

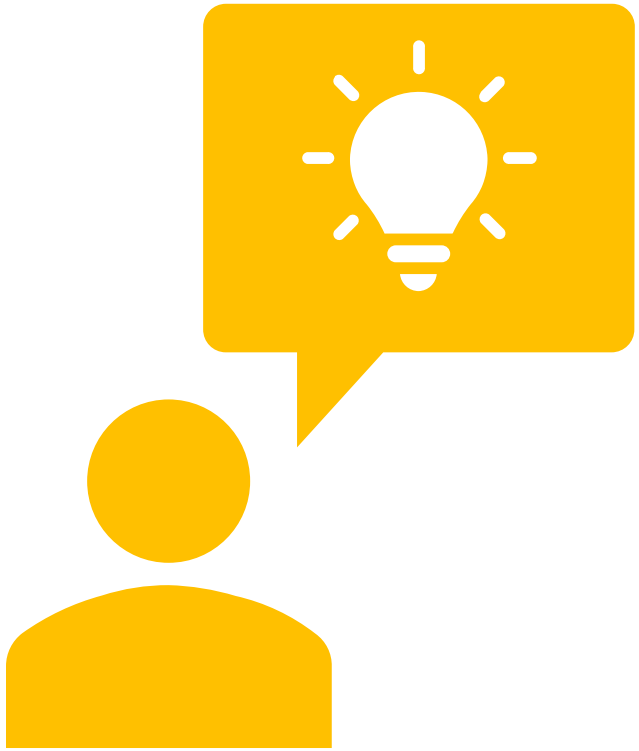
According to the **Part 3**, use a custom in Seaborn:

i. Seaborn v0.8 darkgrid style
ii. 8x5 inch figure
iii. A scatter plot of Math vs Science scores
iv. Custom styling: darkred color and 's' (square) marker
v. Add labels for each point

Output:



KNOWLEDGE CHECK



THANK YOU



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